

National University of Science & Technology

Institute of Geographical Information Systems

Course Title: Digital Mapping and Image Processing

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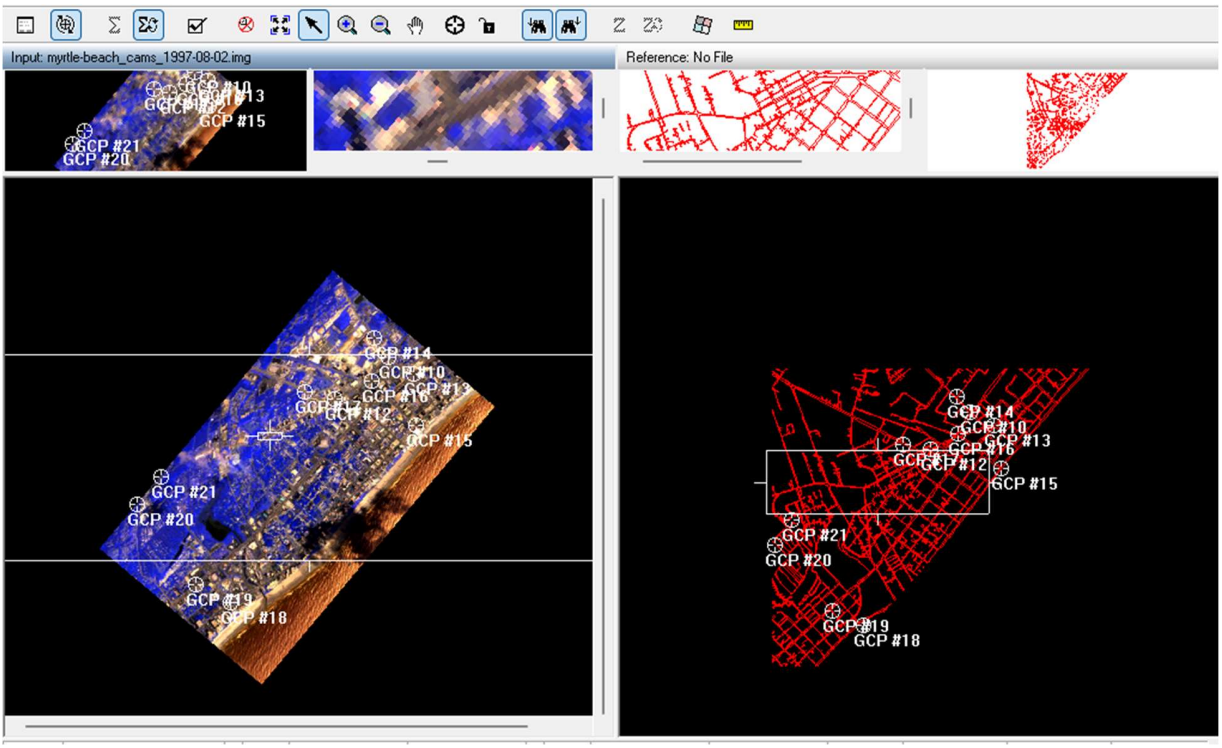


Figure 1 overview of taken GCPs

Point #	Point ID	>	Color	X Input	Y Input	>	Color	X Ref.	Y Ref.	Type	X Residual	Y Residual	RMS Error
1	GCP #10			230.433	-70.930			696250.710	3730532.612	Control	0.123	-0.051	0.13
2	GCP #12			198.795	-228.097			695895.129	3730200.872	Control	0.114	-0.095	0.14
3	GCP #13			303.401	-63.612			696459.295	3730410.708	Control	-0.135	0.094	0.16
4	GCP #14			176.498	-57.393			696128.693	3730670.576	Control	0.121	-0.103	0.15
5	GCP #15			383.099	-152.310			696521.118	3730035.516	Control	-0.277	0.205	0.34
6	GCP #16			237.028	-141.867			696146.229	3730342.849	Control	0.017	-0.021	0.02
7	GCP #17			132.276	-259.714			695661.864	3730249.893	Control	0.256	-0.203	0.32
8	GCP #18			317.283	-754.333			695308.912	3728657.113	Control	-0.009	0.010	0.01
9	GCP #19			228.256	-773.998			695035.665	3728779.348	Control	0.176	-0.133	0.22
10	GCP #20			1.057	-715.776			694524.746	3729362.324	Control	0.625	-0.479	0.78
11	GCP #21			3.172	-630.208			694675.356	3729578.391	Control	-1.012	0.775	1.27
12	GCP #22	▶				▶				Control			

Figure 2 Status of taken GCP

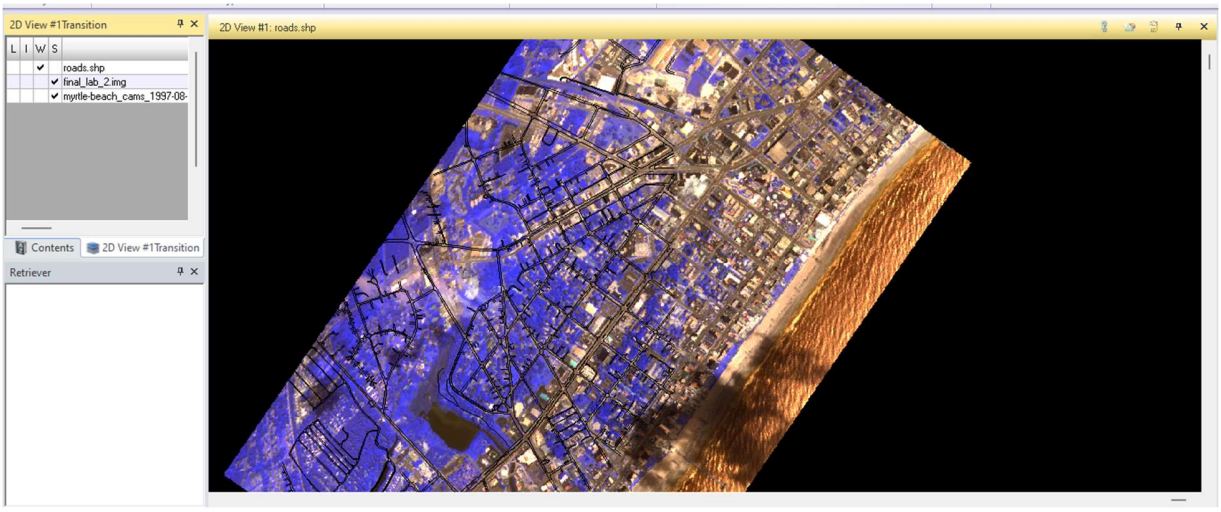


Figure 3 Accuracy Check

Image Rectification Lab Answers

Question 1: Describe the difference between the two image rectification techniques you used and the associated errors that might be introduced when using each of these techniques.

> **Image-to-map rectification** - uses real-world coordinates & GCPs to transform images to standard map projections; errors include coordinate measurement mistakes & projection distortions • **Image-to-image registration** - aligns one image to another without map coordinates; errors from feature matching problems & scale differences • **Usage** - use image-to-map for precise geographic positioning, use image-to-image for quick comparisons when map coordinates aren't needed

Question 2: Is the spatial distribution of your GCPs important to ensure acceptable rectification? Why or why not?

> **Yes, super important** - evenly distributed GCPs across entire image area ensure uniform geometric correction throughout • **Clustered GCPs suck** - putting all GCPs in one area leads to poor rectification & higher RMS errors in uncovered regions •

Bad distribution = bad results - causes geometric distortions & unreliable rectification outside the GCP coverage area

Question 3: What is your definition of the ideal Ground Control Point (GCP)?

Clear & identifiable - must be easily seen in both source image & reference map, like road intersections or building corners • **Permanent & stable** - features that don't change over time, have high contrast with surroundings for precise location • **Small & accurate** - point-like features rather than large areas to minimize location uncertainty & provide accurate coordinates